

Total No. of Questions : 10]

SEAT No. :

P2461

[Total No. of Pages : 2

[5253] - 184

T.E. (Computer Engg.)

DATABASE MANAGEMENT SYSTEMS APPLICATIONS

(2012 Course) (End Semester)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q. 10
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary

Q1) a) Explain View and Index Objects in SQL. [5]

b) Draw an ER diagram for Banking Enterprise. [5]

OR

Q2) a) Explain Normalization. Explain 3 NF with Example. [5]

b) Explain advantages of DBMS over file systems. [5]

Q3) a) Consider following structure for MongoDB collections and write a query for following requirements in MongoDB (Any 3) [5]

Teachers(Tname, dno, experience, salary, date_of joining)

Students(Sname, roll_no, class)

- i) Write a MongoDB query to create above collections & for insertion of some sample documents.
- ii) Find the information about all teachers of dno = 2 and having salary greater than or equal to 10,000/-
- iii) Find the student information having roll_no = 2 or Sname = Anil
- iv) Display Total no of Students of TE Class

b) Explain key-value data store model of NOSQL databases. [5]

OR

Q4) a) What are NOSQL Database Types? Explain Document database type in detail. [5]

b) What are ACID Properties of Transaction? Explain Isolation in detail. [5]

P.T.O.

- Q5) a)** Explain different Parallel System Architectures with their advantages and disadvantages. [8]
b) Explain 3-tier and 2-tier web architecture with diagram considering suitable example. [8]

OR

- Q6) a)** Explain different steps required for JAVA to MongoDB database connection using JDBC. [8]
b) Explain distributed database system architecture. [8]

- Q7) a)** What is XML? Explain XQuery and FLWOR with Example. [7]
b) Explain Hadoop Architecture in Detail. Also explain how map reduce works. [10]

OR

- Q8) a)** Write XML Document for Book Data (Category, Title, Author, and Price). Write XQuery to retrieve all book information with price>30. [7]
b) Write a Short note on R Programming [5]
c) Write a Short note on Hive Database [5]

- Q9) a)** Explain BIS Components in Detail. [5]
b) Explain Association Rule mining Algorithm with Example [7]
c) Differentiate between OLAP and OLTP. [5]

OR

- Q10) a)** Write short note on Data-mining regression analysis. [5]
b) Explain K Means Clustering Algorithm with Suitable Example. [7]
c) What is Classification? Explain different steps in Classifications? [5]

